

Measurement-Based Model for Cross-Absorption Modulation in an MQW Electroabsorption Modulator

N. Cheng and J. C. Cartledge, *Senior Member, IEEE, Member, OSA*

Abstract—A model for cross-absorption modulation in a multiple-quantum-well electroabsorption modulator is developed based on measurements of the escape time of photogenerated carriers, the dependence of the modulator absorption on the power and wavelength of the pump and probe signals, and the dependence of the α parameter of the wavelength-converted signal on the power and wavelength of the pump and probe signals. Good agreement between calculated and measured results is demonstrated for static, small-signal and large-signal conditions. The accuracy and computational efficiency of the model in describing the intensity and phase modulation properties of a wavelength-converted signal make it suitable for computer simulations aimed at system design and performance evaluation.

Index Terms—Cross-absorption modulation, electroabsorption (EA), wavelength conversion.

I. INTRODUCTION

ELECTROABSORPTION modulators (EAMs) exhibit nonlinear optical transmission characteristics that make them suitable for all-optical wavelength conversion and regeneration based on cross-absorption modulation [1]–[4]. To effectively assess the implications of the device properties on system performance, models that accurately describe the static and dynamic properties are needed. Detailed models of cross-absorption modulation in an EAM provide a fundamental understanding of device properties but are generally too complicated and computationally intensive for computer simulations aimed at system design and performance evaluation [5], [6]. The values of the parameters in these device models can be difficult to determine precisely and can vary for nominally identical devices due to fabrication tolerances. For system design and performance evaluation purposes, computationally efficient device models are needed that accurately represent the intensity and phase modulation properties of the output signal. A useful approach is to develop models that use device measurements to obtain the required relationships between parameters. Such models must achieve an appropriate compromise between the complex device properties that need to be considered and the requirement that the models be computationally efficient for system performance studies without compromising accuracy.

For EAMs used as optical modulators (electrical-to-optical, or E/O), the measured dependence of the absorption and α parameter on the bias voltage has been used to develop models for the static and dynamic device properties [7], [8].¹ This paper presents a similar approach for EAMs used as all-optical wavelength converters and regenerators (optical-to-optical, or O/O). The method for modeling cross-absorption modulation in a multiple-quantum-well (MQW) EAM is based on the following device measurements: the carrier escape time from the quantum wells; the dependence of the modulator absorption on the power and wavelength of the pump and probe signals; and the dependence of the α parameter of the wavelength-converted signal on the power and wavelength of the pump and probe signals. In addition to the requisite change in the stimulus from an electrical signal to an optical signal, the method includes an important extension to the approach that has been used for optical modulators (E/O); the time dynamics are explicitly considered in order to capture the finite bandwidth of the cross-absorption-modulation process. This is achieved by using a rate equation for the number of photogenerated carriers and expressing the measured results for the fiber-to-fiber loss and α parameter in terms of the number of carriers. Using this measurement-based model, good agreement is demonstrated between calculated and measured results for the intensity and time-resolved chirp of a wavelength-converted signal.

II. STATIC RESULTS

For cross-absorption between pump and probe signals, the rate equation for the number of photogenerated carriers in the quantum wells depends on the absorption of the two signals and is given by

$$\frac{dN}{dt} = \frac{10^{-\Gamma_c/10}}{hc} \left[\left(1 - 10^{-(\Gamma(\lambda_{pu}, N) - 2\Gamma_c)/10} \right) P_{in,pu}(t) \lambda_{pu} + \left(1 - 10^{-(\Gamma(\lambda_{pr}, N) - 2\Gamma_c)/10} \right) P_{in,pr} \lambda_{pr} \right] - \frac{N}{\tau} \quad (1)$$

where N is the number of carriers, Γ_c is the input and output coupling loss in decibels, Γ is the fiber-to-fiber loss in decibels, P_{in} is the power of the input signal, hc/λ is the photon energy, and τ is the carrier escape time. Throughout the paper,

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The authors are with the Department of Electrical and Computer Engineering, Queen's University, Kingston, ON K7L 3N6, Canada.

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¹Several measurement techniques are known for determining the dependence of the α parameter on the bias voltage [9]–[14].

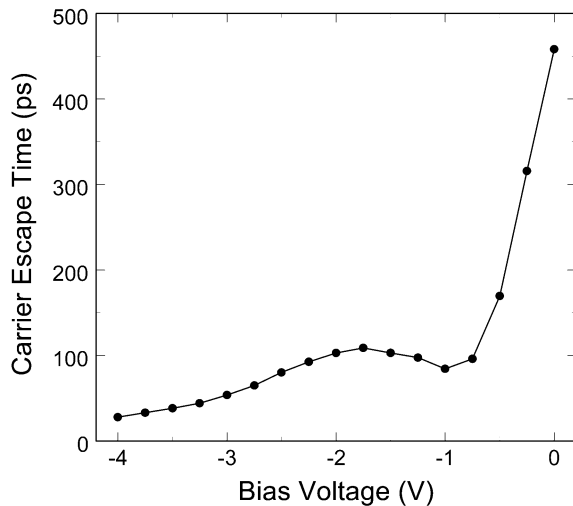


Fig. 1. Dependence of the carrier escape time on the bias voltage.

the subscripts pu and pr denote the pump and probe signals, respectively. The fiber-to-fiber loss depends on the wavelength of the signal and the number of photogenerated carriers. The first term on the right-hand side (RHS) of (1) (in brackets) represents the generation of carriers due to the absorption of the pump and probe signals. It is written in terms of the coupling loss and fiber-to-fiber loss, which are measurable quantities. The second term on the RHS represents the loss of carriers due to escape from the quantum wells. In (1), it is assumed that absorption leads to carrier generation in the quantum wells (scattering and radiation losses are negligible), the input and output coupling losses are the same, and the carrier escape time is independent of the number of carriers (as justified hereafter by measurements of the escape time). In general, (1) is solved numerically for time-varying pump signals. Analytical solutions are obtained for a continuous-wave (CW) pump signal and for a sinusoidally time-varying pump signal under small-signal conditions.

The MQW EAM used to illustrate the measurement-based model has InAlGaAs wells and InAlAs barriers [15]. The well layers are tensile strained for low polarization sensitivity. The device is intended for optical modulation (E/O) and is not optimized for wavelength conversion or optical regeneration based on cross-absorption modulation (O/O) [4], [6]. The carrier escape time τ is determined experimentally from the optical-to-electrical frequency response, as obtained with a lightwave component analyzer [16]. The dependence of the carrier escape time on the bias voltage is shown in Fig. 1. The wavelength and average power of the input optical signal are 1551.72 nm and -15 dBm, respectively. The carrier escape time decreases from 458 to 28 ps as the bias voltage decreases from 0 to -4 V since the applied electric field enhances the carrier sweep-out. Similar results were obtained for average input powers up to 10 dBm and for other wavelengths. The coupling loss Γ_c is estimated as 4.2 dB/facet from the insertion loss for a wavelength of 1600 nm, which is far above the band edge of the quantum wells.

The dependence of the fiber-to-fiber loss Γ on the input signal power is illustrated in Fig. 2 for wavelengths of 1550.12 and 1551.72 nm and for different values of the bias voltage. As

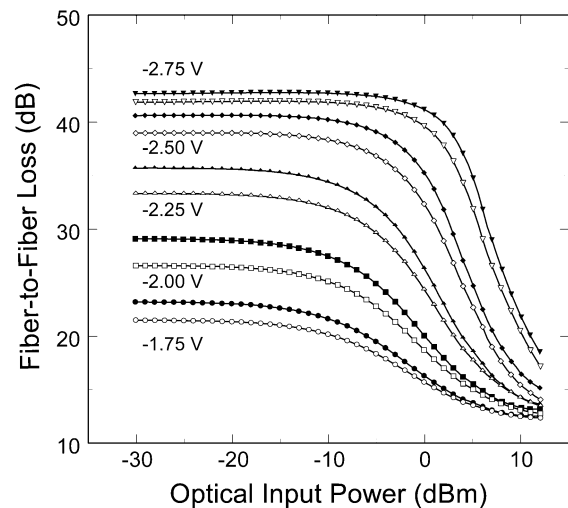


Fig. 2. Dependence of the fiber-to-fiber loss on the input signal power for different values of the bias voltage. Wavelength is 1550.12 nm for the solid symbols and 1551.72 nm for the open symbols.

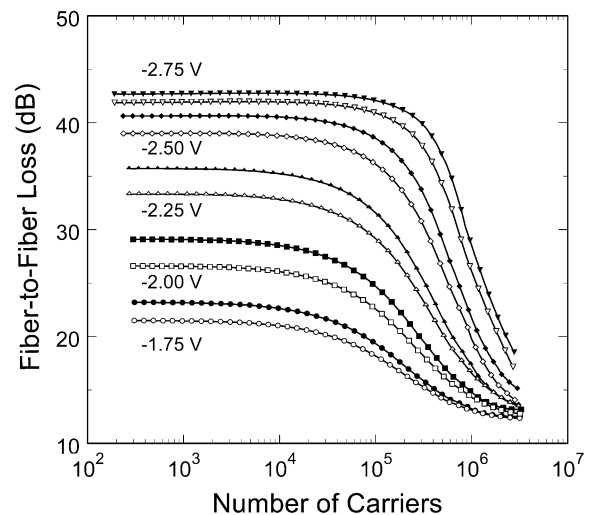


Fig. 3. Dependence of the fiber-to-fiber loss on the number of photogenerated carriers for different values of the bias voltage. Wavelength is 1550.12 nm for the solid symbols and 1551.72 nm for the open symbols.

the input power increases, the photogenerated carriers in the quantum wells cause the absorption to decrease due to the band-filling effect and the space charge field. As the bias voltage decreases (becomes more negative), the decrease in the carrier escape time leads to a decrease in the number of carriers in the quantum wells and a corresponding increase in the power required to saturate the absorption. Using the carrier escape time and $N = P_{\text{abs}}\lambda\tau/hc$, where P_{abs} is the absorbed power, Fig. 3 illustrates the measured results for the fiber-to-fiber loss in terms of the number of photogenerated carriers for use in (1).

Measured results (symbols) are shown in Fig. 4 for the cross-absorption with different values of the reverse-bias voltage. The power and wavelength of the probe signal are 0 dBm and 1550.12 nm, respectively. The wavelength for the copropagating pump signal is 1551.72 nm. Due to the absorption saturation, an increase in the pump power reduces

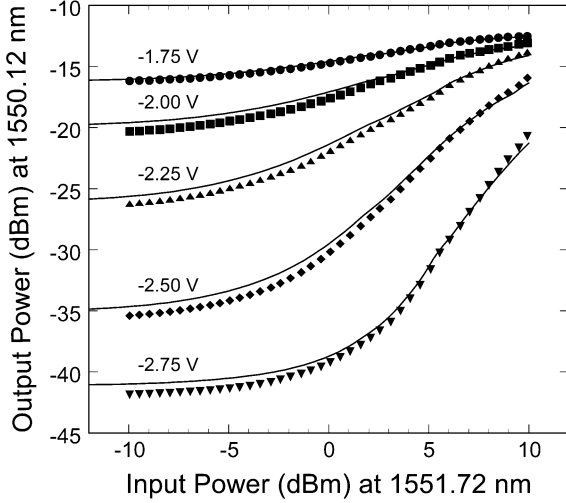


Fig. 4. Cross-absorption between pump and probe signals with wavelengths of 1551.72 and 1550.12 nm for different bias voltages: measured results are denoted by symbols, and calculated results are denoted by lines.

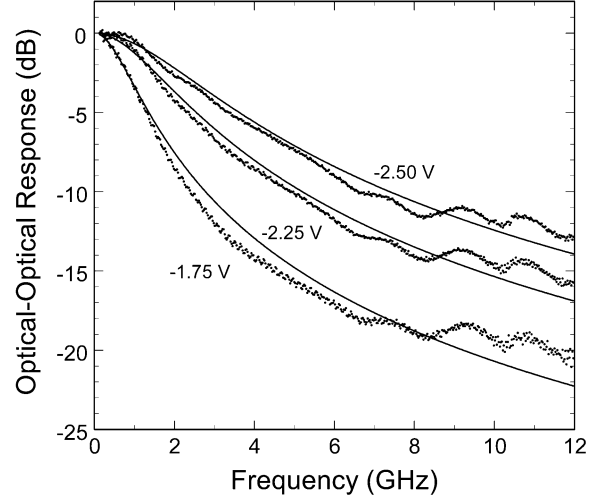


Fig. 5. Frequency response of optical-to-optical conversion due to cross-absorption modulation for bias voltages of -1.75 , -2.25 , and -2.5 V: measured results are denoted by symbols, and calculated results are denoted by lines.

the modulator absorption and, hence, increases the output probe power. The steady-state version of (1) can be written as

$$N = \frac{\tau 10^{-\Gamma_c/10}}{hc} \left[\left(1 - 10^{-(\Gamma(\lambda_{pu}, N) - 2\Gamma_c)/10} \right) P_{in,pu} \lambda_{pu} + \left(1 - 10^{-(\Gamma(\lambda_{pr}, N) - 2\Gamma_c)/10} \right) P_{in,pr} \lambda_{pr} \right]. \quad (2)$$

Since the RHS of (2) is a monotonically decreasing function of N (see Fig. 3), a unique solution exists for the number of photogenerated carriers. The solution to (2) is obtained using the bisection method. In solving (2), spline interpolation is used to determine the values of the fiber-to-fiber loss between measured data points. Once the carrier number is known, the fiber-to-fiber loss $\Gamma(\lambda_{pr}, N)$ is determined from the results in Fig. 3 using spline interpolation, and the output probe power is then given by

$$P_{out,pr} = P_{in,pr} 10^{-\Gamma(\lambda_{pr}, N)/10}. \quad (3)$$

The calculated results for the cross-absorption saturation are also shown in Fig. 4 (lines) and exhibit good agreement with the measured results.

III. SMALL-SIGNAL RESULTS

The small-signal optical-to-optical intensity modulation frequency response is measured using a lightwave component analyzer. The power and wavelength of the probe signal are 0 dBm and 1550.12 nm, respectively. The power and wavelength of the copropagating pump signal are 0 dBm and 1551.72 nm, respectively. At the output of the EAM, an optical filter is used to select the output probe signal. Calibrated results for the frequency response are shown in Fig. 5 for different values of the bias voltage. The responses are normalized to the value at the

lowest modulation frequency (130 MHz). For sinusoidal intensity modulation of the pump signal at a frequency ω_m

$$P_{pr} = P_{pr,0} \quad (4)$$

$$P_{pu} = P_{pu,0} + P_{pu,1} \exp(j\omega_m t) \quad (5)$$

$$N = N_0 + N_1(\omega_m) \exp(j\omega_m t) \quad (6)$$

the rate equation (1) is solved analytically for small-signal conditions ($P_{pu,1} \ll P_{pu,0}$). Assuming that the change in the fiber-to-fiber loss is proportional to the change in the carrier number, the normalized frequency response for the optical-to-optical conversion is given by

$$H(\omega_m) = \frac{1}{1 + j\omega_m \tau_{eff}} \quad (7)$$

where the effective time constant τ_{eff} is

$$\tau_{eff} = \left(\frac{1}{\tau} - \frac{\xi_{pr} P_{pr,0} \lambda_{pr}}{hc} - \frac{\xi_{pu} P_{pu,0} \lambda_{pu}}{hc} \right)^{-1}. \quad (8)$$

Here

$$\xi = \left. \frac{d\kappa}{dN} \right|_{N=N_0} \quad (9)$$

and κ is the ratio of the absorbed optical power to the input power

$$\kappa = 10^{-\Gamma_c/10} \left(1 - 10^{-(\Gamma(\lambda, N) - 2\Gamma_c)/10} \right). \quad (10)$$

This yields

$$\xi = 10^{-(\Gamma(\lambda, N) - \Gamma_c)/10} \left. \frac{\ln 10}{10} \frac{d\Gamma(\lambda, N)}{dN} \right|_{N=N_0} \quad (11)$$

where $d\Gamma(\lambda, N)/dN$ is determined from Fig. 3. Since ξ is a small number and $P_0 \lambda / hc$ is a large number, it is difficult to accurately calculate τ_{eff} directly from (8) given the uncertainty

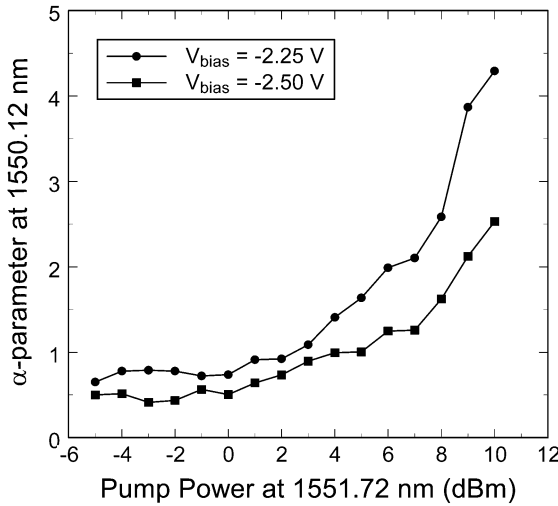


Fig. 6. Dependence of the α parameter on the power of the input signal for bias voltages of -2.25 and -2.5 V.

in determining $d\Gamma(\lambda, N)/dN$ from experimental data. Therefore, the calculated results for the amplitude of the frequency response $|H(\omega_m)|$ shown in Fig. 5 use the effective time constant as a fitting parameter. For bias voltages of -1.75 , -2.25 , and -2.5 V, the effective time constant is 172, 93, and 65 ps, respectively. As the reverse-bias voltage increases, the carrier escape time decreases and the effective time constant decreases. This leads to an increase in the -3 -dB bandwidth. For this particular EAM, the carrier escape time leads to a bandwidth of 4.3 GHz for a bias voltage of -2.5 V. The correspondence between the calculated and measured responses provides an indication that (1) is an appropriate description for the number of photogenerated carriers.

The large-signal model uses the dependence of the small-signal α parameter on the number of photogenerated carriers to characterize the chirp of the wavelength-converted signal. Sinusoidal intensity modulation of the pump signal causes both the intensity and phase of the wavelength-converted signal (output probe signal) to be modulated (Kramers–Krönig relation). The corresponding frequency chirp is characterized by the α parameter [7]

$$\alpha = \frac{\Delta n_r}{\Delta n_i} \quad (12)$$

$$= 2 \frac{d\phi}{d(\ln I)} \quad (13)$$

where Δn_r and Δn_i are the changes in the real and imaginary components of the complex refractive index due to the pump signal, respectively, and ϕ and I are the phase and intensity of the wavelength-converted signal, respectively. For an EAM, the material expression (12) and signal expression (13) for the α parameter are equivalent. The α parameter of the wavelength-converted signal is measured using the fiber frequency response technique [10]. The power and wavelength of the probe signal are 0 dBm and 1550.12 nm, respectively. The wavelength for the copropagating pump signal is 1551.72 nm. For bias voltages of -2.25 and -2.5 V, results are presented in Fig. 6 as a function of the pump power and in Fig. 7 as a function of

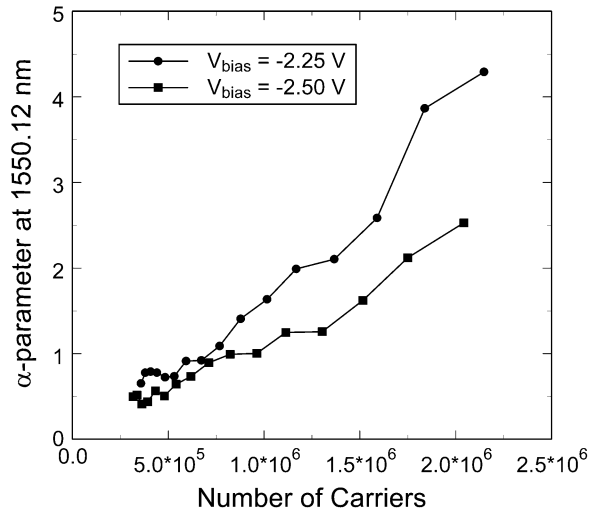


Fig. 7. Dependence of the α parameter on the number of photogenerated carriers for bias voltages of -2.25 and -2.5 V.

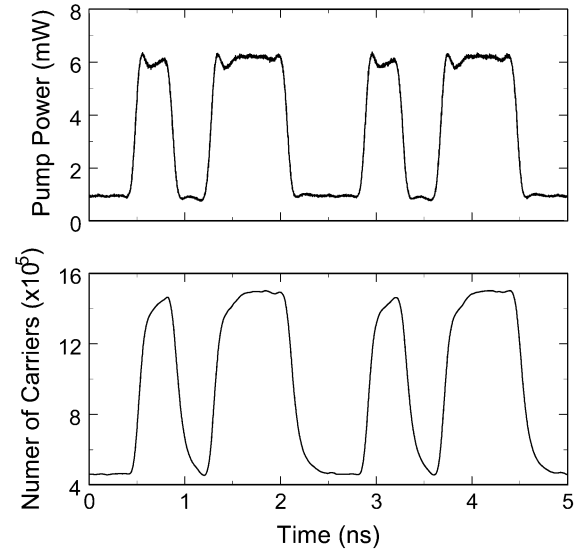


Fig. 8. Pump signal for wavelength conversion and the solution to (1) for the number of photogenerated carriers $N(t)$.

the number of photogenerated carriers. Both the α parameter and cross-absorption (Fig. 4) exhibit a stronger dependence on the pump power for powers larger than 0 dBm compared with powers less than 0 dBm.

IV. WAVELENGTH CONVERSION

From these results, it is evident that the dynamics of the photogenerated carriers must be considered to accurately predict the time dependence of the wavelength-converted signal (intensity and phase). When the pump signal is an intensity-modulated digital waveform, the nonlinear differential equation (1) is solved numerically using a fourth-order Runge–Kutta method. Spline interpolation is again used to determine the values of $\Gamma(\lambda, N)$. In order to compare calculated and measured results for the wavelength-converted signal, the measured pump signal is used in solving (1). For an input probe power of 0 dBm and an average input pump power of 6 dBm, the pump signal and the corresponding solution for $N(t)$ are shown in Fig. 8. Using

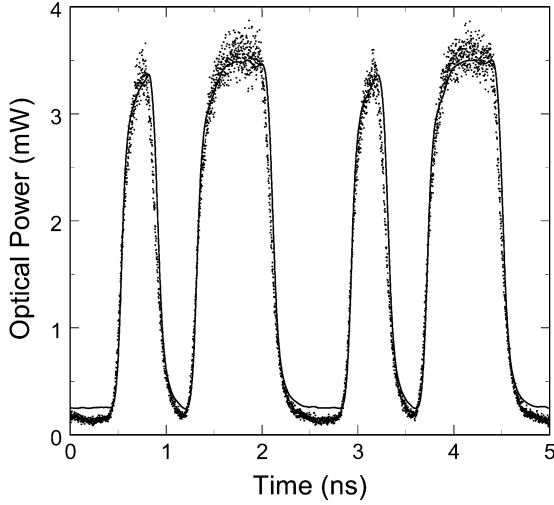


Fig. 9. Calculated (line) and measured (symbols) results for the power of the wavelength-converted signal.

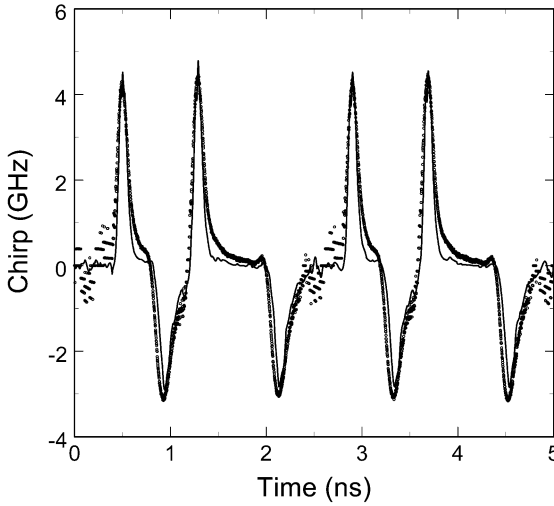


Fig. 10. Calculated (line) and measured (symbols) results for the time-resolved chirp of the wavelength-converted signal.

$N(t)$, the time variation of the fiber-to-fiber loss and α parameter are determined from the results in Figs. 3 and 7. Finally, the power of the wavelength-converted signal $P_{\text{out,pr}}$ is obtained using (3), and the chirp of the wavelength-converted signal is obtained using

$$\begin{aligned} \Delta\nu(t) &= \frac{1}{2\pi} \frac{d\phi}{dt} \\ &= \frac{\alpha(t)}{4\pi} \frac{d \ln(P_{\text{out,pr}}(t))}{dt} \end{aligned} \quad (14)$$

from which the phase is obtained. The desired result for the wavelength-converted signal is then given by

$$E_{\text{out,pr}}(t) = \sqrt{P_{\text{out,pr}}(t)} \exp(j\phi(t)). \quad (15)$$

The results for the power and chirp are shown in Figs. 9 and 10, respectively, for conversion from a wavelength of 1551.72 to

1550.12 with a bias voltage of -2.5 V. For a 2.5-Gb/s signal the transient chirp at the rising and falling edges of a pulse is clearly distinguished. Figs. 9 and 10 also illustrate corresponding measured results. The output signal from the EAM is amplified and filtered before being measured, and the net gain of the amplifier-filter combination is used in obtaining the numerical results. The time-resolved chirp is measured using a Fabry-Pérot filter (-3 -dB bandwidth of 54.5 GHz) as an optical frequency discriminator [17]. A good agreement between the measured and calculated results has also been obtained for other values of the bias voltage and for conversion to other wavelengths. In practice, the wavelength range is limited by an increase in the absorption for smaller values of the probe wavelength and by a decrease in the extinction ratio of the wavelength-converted signal for larger values of the probe wavelength. This method for modeling wavelength conversion in EAMs can also be applied to devices with a smaller carrier escape time which allows for higher bit-rate operation.

The agreement in Figs. 9 and 10 demonstrates the usefulness of the model for the purposes of evaluating system performance. For example, the intensity and phase modulation properties of an optical signal play an important role in assessing the impact of fiber transmission properties (group velocity dispersion and nonlinear effects) on system performance. Moreover, the large number of parameters involved in the system design process places considerable emphasis on the computational efficiency and accuracy of device models.

V. SUMMARY

A method based on measured characterization data is described for modeling wavelength converters that use cross-absorption modulation in an EAM. Good agreement is demonstrated between calculated and measured results for the intensity and time-resolved chirp of the wavelength-converted signal. The model is intended for evaluating system performance where both computational efficiency and accuracy are important.

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N. Cheng received the B.Eng. degree in electronic engineering from Xi'an Jiaotong University, Xi'an, China, in 1992, the M.Eng. degree in electronic engineering from Southeast University, Nanjing, China, in 1995, and the Ph.D. degree in electrical engineering from Queen's University, Kingston, ON, Canada, in November 2003.

He is currently a Postdoctoral Fellow in the Lightwave Systems Research Laboratory, Department of Electrical and Computer Engineering, Queen's University. His interests include photonic technologies and optical communications.

J. C. Cartledge (S'74–M'79–SM'96) received the B.Sc. degree in mathematics and engineering and the M.Sc. and Ph.D. degrees in mathematics from Queen's University, Kingston, ON, Canada, in 1974, 1976, and 1979, respectively.

From 1979 to 1982, he was a Member of the Scientific Staff at Bell-Northern Research, Ottawa, ON, Canada, where his work involved fiber-optic systems for the exchange access network and high-capacity digital radio systems. Since 1982, he has been with the Department of Electrical and Computer Engineering, Queen's University. He has spent one-year sabbatical leaves with the Lightwave Systems Technology Research Division of Bellcore, Red Bank, NJ, in 1988–1989 and with the Optical Communications Department of Tele Danmark Research, Hørsholm, Denmark in 1995–1996. He has served as a consultant and instructor in the area of lightwave technology to several organizations. His current research interests include optical modulators, optical regenerators, semiconductor lasers, fiber-Bragg-grating-based components, and wavelength converters.

Dr. Cartledge was appointed an Inaugural Recipient of a Queen's Research Chair in recognition of his research contributions to lightwave technology in July 2002. His conference organization activities include serving on the Technical Program Committee for the Conference on Optical Fiber Communications (1994–1997, 2005), the Technical Program Committee for the joint meeting of the International Conference on Integrated Optics and Optical Fibre Communications and the European Conference on Optical Communications (IOOC/ECOC'97), and the Technical Program Committee for the IEEE Lasers & Electro-Optics Society (LEOS) Annual Meeting (2002–2004). He is a Member of the Optical Society of America (OSA).